

Paco Pascal

Programmer, Software Engineer, Craftsman

914.299.4059
me@pacopascal.com
github.com/FragmentedCurve
www.pacopascal.com

Experience

■ IBM

1/2020 → 4/2021

EDA Software Engineer

- Maintained and refactored a 20-year-old licensing component written in C/C++.
- Rebuilt and optimized the broken build system to support RHEL 7.
- Engineered a RESTful API in Go for analytics data collection and reporting.
- Integrated various backends (luci, ThisFile, LDAP) with the API.
- Initiated and contributed to OpenShift and Docker container deployments for the EDA Toolkit.
- Engineered a custom container stack to integrate with OpenAFS and other distributed file systems for EDA needs.
- Gave a formal proof on an IT ticket to show IBM wasn't the cause of a bug in production.
- Assisted IBM Systems and IBM Research teams with containerizing applications.
- Assisted a zPDT colleague with patching the CKD volume verification tool.

■ Yorktown Love in Action

10/2016 → 12/2020

IT & Software Engineer

- Reverse engineered CCTV Windows client to disable UAC.
- Met YLIA needs by patching darkhttpd with HTTPS forwarding support. (Later merged upstream.)
- Server & workstation maintenance.
- TCP/IP Networking (UniFi, OpenVPN, subnet divisions, etc.)
- Deployed and maintained FreeBSD Samba servers providing Active Directory & CIFS for Windows.

Operations

- Designed & built enclosures and apparatuses for the showroom floor.
- Created and implemented the donation intake system which employees and volunteers worked in.
- Fulfilled and organized customer deliveries and pickups.

■ Bitty Bytes

9/2015 → 9/2016

Lead Developer

- Developed a "selling engine" that automated the online listing of liquidated items, significantly increasing sales efficiency and volume.
- Engineered a data integration system, combining data from IoT devices, human input and APIs to optimize pricing and enhance search engine optimization (SEO).
- Used Python to develop our image processing application in Python.
- Worked with SQLite and Postgres as the DB for our backend that was also built in Python.
- Scouted and built relationships with material suppliers.
- Integrated hardware and software solutions to be ergonomic for employees to achieve maximum productivity.
- Conducted empirical tests on methodologies and solutions to steer the company's success forward.
- Spearheaded technologies best suited for the project.
- Hired workers and volunteers.
- Found a worker's best fit in the production line.

■ BitBison LLC

12/2013 → 6/2014

Software Engineer

- Architected and built a platform for managing large mining-rigs.
- Solved issues with how to automate the stateless configuration of miners.
- Maintained a custom live Linux distro derived from Debian & Finnix.
- Developed and maintained an init script codebase responsible for automating the initialization onto a grid of worker miner nodes.
- Responsible for rolling external modules and patches into the distro's kernel.

Projects

- **FreeBSD**
 - Mostly submitting patches for userland bug fixes (sh, elftoolchain, etc.).
 - Port maintainer.
- **Arch Linux AUR Maintainer**
 - Managed a collection of Guile Scheme and miscellaneous packages.

Skills

- **Programming Languages** C C++ Go Python Awk Lisp Bash x86 Assembly
- **Version Control Software** Git SVN
- **Compilers & Debuggers** GCC GDB clang
- **Operating Systems** FreeBSD Linux RHEL (Red Hat Enterprise Linux)
- **Containers & VMs** Docker KVM QEMU Virtual Box
- **Databases** Postgres MariaDB SQLite3
- **IaaS & PaaS** AWS (ECS, EC2) Heroku
- **Software Development** RESTful API Development Containerization and Deployment Build System Optimization Reverse Engineering

Education

Mathematics, AS

SUNY Westchester Community College

Graduated Spring 2012

Computer Science, BSc

Arizona State University

Currently Enrolled

Certifications

Amazon Software Developer

Issued by Amazon (via Coursera)

Software Engineer

Issued by IBM